

How Noisy is Automatic Annotation? Characterizing Label Noise from Generative Models in Multi-Dimensional Spanish Subjectivity Annotation

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Abstract

Although human variability has traditionally been the primary source of annotation noise, the increasing use of Large Language Models (LLMs) for automated labeling is shifting this landscape toward systematic, machine-generated label noise. This challenge is particularly severe in Subjectivity Detection (SD), where the behavior of automated errors remains largely unexamined. This study characterizes the extent and linguistic nature of automated annotation noise in Spanish SD. Errors are evaluated as omissions (under-generation) or additions (over-generation) across six subjectivity dimensions: Doubt, Certainty, Specificity, Non-specificity, Distancing, and Participation. Using an expert-annotated human gold standard as a reference, a traditional machine learning control is benchmarked against three state-of-the-art LLM families (GPT, Gemini, and Llama). The results reveal a clear architectural divide: while the traditional ML framework predominantly generates omission errors (63.99%), only detecting markers it was explicitly programmed to recognize, LLMs like Gemini and Llama lean heavily toward addition errors (exceeding 51%), guessing and hallucinating markers instead of leaving uncertain cases unlabeled. Although Gemini minimizes overall noise, achieving F1-scores up to 0.96 in specific dimensions, Certainty remains highly susceptible to errors, showing a 53.96% error rate in the traditional ML control. Ultimately, this diagnostic evaluation of LLMs as annotators in this specific subjective task reveals that automated systems introduce systematic label noise, demonstrating that human-in-the-loop participation is indispensable to resolve these errors and ensure reliable dataset curation.

1 Introduction

The boundary between natural user-generated text (UGT) and AI-generated content is becoming increasingly blurred. Instead of just analyzing digital

discourse, Large Language Models (LLMs) are now widely used to automatically annotate datasets (Klie et al., 2023). Consequently, data corruption in modern NLP pipelines has transitioned from random human variance into systematic, AI-induced generative label noise (Nasution and Onan, 2024). When LLMs attempt to interpret subjective, informal, or non-standard online text, they frequently introduce this noise.

From a noisy-text perspective, these annotation errors are more than just low performance metrics; they represent a distinct form of LLM-generated noisy text that can compromise both dataset quality and the training of downstream models. This issue is especially clear when annotating natural online discourse, which is highly subjective. Pragmatic and discursive tasks, such as Subjectivity Detection (SD), are widely recognized as some of the most complex annotation challenges, being highly susceptible to errors (Wiebe et al., 2005). Because expressing subjectivity is a highly individual act (Bajtin, 1982), capturing its nuances requires interpreting the speaker’s stance and intent within a specific context (Wilson et al., 2005). When annotating these subtle dimensions, LLMs often struggle with linguistic ambiguity, resulting in systematic errors, particularly in non-English languages like Spanish. Despite the risks of automatic annotation, how this LLM-generated noise behaves and distributes across different subjectivity dimensions remains largely unexamined.

To address this gap, this study presents an exploratory diagnosis of LLM-generated label noise in Spanish subjectivity annotation. As previously shown by Piot et al. (2026), traditional evaluation metrics fall short when assessing LLMs as reliable annotators in subjective tasks. For this reason, instead of relying on broad statistical averages, a granular noise-codification framework is proposed to analyze under-generation (*omissions*) and over-generation (*additions*) in high-density, carefully an-

notated texts. This systematic approach pursues a three-fold objective: i) to assess the viability of automated annotation in these high-density settings; ii) to deploy the proposed diagnostic framework to characterize model-specific error profiles; and iii) to analyze how generative models handle label noise in pragmatically complex discourse.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 introduces the multi-dimensional subjectivity annotation scheme and the proposed noise diagnostic methodology. The experimental setup, including the evaluation dataset and the comparison between traditional and LLM pipelines, is detailed in Section 5. Section 6 discusses the quantitative and qualitative findings regarding AI-induced label noise, and Section 7 concludes the paper and outlines its limitations.

2 Related Work

Historically, text noisiness from a data quality perspective focused primarily on surface-level defects, such as typographical errors and OCR corruption (Bhadauria et al., 2024). However, as modern language models have achieved broad proficiency in general linguistic tasks (Chang et al., 2024), the research focus has shifted toward deeper, more complex tiers of data corruption driven by their deployment as automated, silver-standard annotators (Jahan et al., 2024). Within this paradigm, the primary bottleneck is no longer surface-level accuracy. Instead, the challenge shifts to pragmatic and discursive layers, where the models’ limitations in metalinguistic analysis and complex reasoning lead to systematic errors (Amin et al., 2024).

When LLMs are used for automated annotation, this limitation often results in a clear trade-off, yielding high recall but lower precision (Nasution and Onan, 2024). This vulnerability becomes even more severe when handling highly ambiguous tasks (Voutsas et al., 2025). Even when evaluated across sophisticated linguistic benchmarks (Srivastava et al., 2023; Ettinger et al., 2023), LLMs frequently produce human-like outputs that lack true metalinguistic comprehension (Song et al., 2025), which reveals that models can offer a comparative evaluation path, despite presenting instance-level divergence in complex annotation tasks (Piot et al., 2026). Recent work highlight that traditional evaluation approaches on LLMs as annotators for complex tasks, such as hate speech or cognitive

distortions detection, frequently fall short because the variability performance under subjective conditions is not considered (Piot et al., 2026; Sharma et al., 2026; Xu et al., 2026).

Regarding LLMs as annotators, the lack of metalinguistic comprehension is highly visible in subjectivity analysis, which focuses on evaluating authorial involvement and stance within a message (Kasmuri and Basiron, 2017). At a conceptual level, subjectivity represents the explicit or implicit commitment of the author, carrying rich pragmatic features, including personal opinions and intentions, that depend heavily on context (Hyland, 2005). Because capturing these nuances requires deep inferential reasoning, automated annotation methods struggle to maintain data validity in this domain (Shokri et al., 2024). Evaluating LLMs is therefore necessary to determine if generative architectures can handle such non-surface-level linguistic tasks. Conversely, previous work showed that LLMs can serve as consistent annotators for inherently subjective tasks by reframing the annotation process as a repeated querying rather than a single shot prediction (Sharma et al., 2026).

Besides, this shift toward evaluating generative models contrasts with traditional dataset curation, where Annotation Error Detection (AED) has historically focused on identifying human inconsistencies using structural or lexical patterns (Dickinson and Meurers, 2003; Klie et al., 2023). However, raw quantitative error metrics often mask qualitative failures in nuanced linguistic domains (Barnes et al., 2019). This limitation highlights the need for more granular evaluation methods to ensure dataset quality and scientific validity (Dickinson, 2015).

This generative noise becomes especially prominent in pragmatic tasks where models struggle to interpret implicit markers, causing them to diverge from human gold standards (Schroeder et al., 2025). Crucially, the current literature lacks diagnostic frameworks to characterize the qualitative profiles of the noise introduced by automated annotations (Klie et al., 2023). This study addresses this gap by proposing an exploratory evaluation framework designed to profile and analyze AI-induced label noise in Spanish subjectivity tagging, shifting the analytical focus from simple error detection to granular noise characterization.

3 Multi-Dimensional Subjective Detection Annotation

Grounded in Hyland (2005) and Martín Zorraquino (1999), the annotation scheme employed in this study is built around the notion of *authorial commitment*: the degree to which a speaker presents a proposition as certain, concrete, and personally endorsed, rather than tentative or detached. Linguistic devices that weaken this commitment are known as *hedges* (e.g., the Spanish subjunctive, or adverbs like “*maybe*”), while devices that strengthen it are known as *boosters* (e.g., the imperative mood, or adverbs like “*yes*”). This distinction maps directly onto three conceptual axes of pragmatic variance: certainty, concreteness, and involvement. Rather than isolated categories, each axis functions as a bipolar continuum pairing a low-commitment hedge (LC) against a high-commitment booster (HC) counterpart. This balanced structure yields six distinct dimensions:

- **Degree of Certainty (Doubt vs. Certainty):** Evaluates authorial commitment to truth by contrasting hedging markers, such as the subjunctive mood (*Doubt (LC)*), with boosting indicators of conviction (*Certainty (HC)*).
- **Level of Concreteness (Non-specificity vs. Specificity):** Distinguishes between vague, generalizing terms that hedge accountability (*Non-specificity (LC)*) and boosting referential anchors like named entities (*Specificity (HC)*).
- **Scale of Involvement (Distancing vs. Participation):** Measures the author’s discursive presence by contrasting hedging, impersonal or passive constructions (*Distancing (LC)*) with boosting, direct engagement markers (*Participation (HC)*).

Although traditional pragmatic analysis is typically context-based, this scheme is applied at the word or span level, following previous approaches to subjectivity analysis (Vieira et al., 2020; Antici et al., 2021) that have demonstrated the benefits of this granularity. It is designed as a multi-label framework where annotations target specific linguistic markers that are the concrete materialization of the subjectivity dimensions described above. Because a single word or phrase can perform multiple pragmatic functions simultaneously, these labels can freely overlap on the same text span. For instance, the Spanish imperative verb form “*comprad*” (buy) simultaneously triggers two

high-commitment (booster) labels: *Certainty (HC)*, as the imperative mood projects communicative conviction and authority, and *Participation (HC)*, since addressing the audience directly acts as an explicit engagement marker. This multi-label density is essential for capturing the complex pragmatic footprint of natural discourse.

These dimensions are mapped using specific linguistic markers where LLM annotations diverge from human standards (Table 1). Rather than measuring grammatical frequencies, linguistic features, such as verbal moods or pronouns, are treated strictly as vehicles for the underlying pragmatic dimensions of hedging and boosting. Concrete examples of these overlapping markers and their resulting errors are detailed in Section 4 and Appendix A.

Dimension	Linguistic Marker / Trigger
Doubt	Question marks Conditional tense Subjunctive mood Verbs of belief Stressed interrogative pronouns Conditional adverb “ <i>si</i> ” Adverbs indicating possibility
Certainty	Adverb “ <i>si</i> ” Verbs of knowledge Verb “ <i>ser</i> ” (to be) Verb “ <i>estar</i> ” (to be) Verb “ <i>tener</i> ” (to have) Imperative mood Future tense Periphrases of obligation
Specificity	Numbers and digits Proper names and Named Entities Technical terms
Non-specificity	Generalizing terms Indefinite pronouns Temporal constructions
Participation	First-person verbs Second-person verbs First-person pronouns Second-person pronouns
Distancing	Impersonal reflexive passive Passive voice Third-person possessive pronouns

Table 1: Granular taxonomy of linguistic markers across the six subjectivity dimensions. These structural features serve as the target triggers for identifying and evaluating automated annotation noise.

4 Diagnostic Framework for Label Noise Induced by Generative Models

This section introduces a diagnostic framework to evaluate generative label noise by analyzing discrepancies between automated predictions and the human gold standard (Section 5.2). At its core, this framework operates on a binary codification where each predicted label is classified as either correct or incorrect. The analysis focuses specifically on the incorrect predictions, categorizing them into two primary noise profiles that directly map to standard evaluation metrics:

- **Under-generation (*omissions*):** Occurs when the model fails to detect an existing pragmatic marker (False Negatives), representing the qualitative root of degraded Recall performance.
- **Over-generation (*additions*):** Occurs when the model predicts a marker where none exists in the text (False Positives), representing the qualitative driver behind lower Precision scores.

It is worth clarifying that, although the six subjectivity dimensions are a fixed label set, the words or text spans models must extract for each label are not predefined, so models can generate tokens absent from the source text. When this happens, the predicted span does not exist in the source and is counted as an addition (False Positive), regardless of its assigned label. This may indicate hallucination, but characterizing it falls outside this framework’s scope, which focuses only on whether errors manifest as omissions or additions.

Table 2 illustrates an operational example of how these alignments and divergencies are identified.

While standard evaluations rarely classify synthetic data errors using detailed taxonomies (Klie et al., 2023), this methodology goes further by linking each omission and addition to the specific syntactic or pragmatic marker that triggered it, as detailed in Table 1. This word- and span-level mapping enables a highly specialized diagnostic of Spanish subjectivity detection.

This omission/addition framework was not adopted from an external taxonomy, but developed for this study by mapping the observed annotation discrepancies directly onto standard evaluation metrics. This binary error codification follows established practice in annotation error detection (Dickinson and Meurers, 2003; Klie et al., 2023),

Linguistic Marker	Gold Standard	Model Output	Result
Verb of belief	<i>sospecha</i> (suspects)	<i>sospecha</i> (suspects)	Match
Conditional	<i>podría</i> (could)	<i>podría</i> (could)	Match
Subjunctive	<i>hubiera</i> (had been)	<i>hubiera</i> (had been)	Match
Non-marker	X	<i>pudieron</i> (they could)	Addition
Subjunctive	<i>hagan</i> (they do)	X	Omission
Conditional	<i>deberían</i> (should)	X	Omission

Table 2: Examples of error classification for the *Doubt* dimension. The table shows how linguistic triggers are classified as a match, omission, or addition when compared against the gold standard.

which favors empirically grounded error categorization over fixed *a priori* schemes when diagnosing task-specific noise.

Although this marker-level taxonomy is tailored to subjectivity analysis, the underlying diagnostic framework remains highly adaptable: since it fundamentally relies on identifying matches, omissions, and additions at the word or span level, it can be extended to other multi-label tagging tasks by simply mapping the *omission/addition* categories to task-specific features.

5 Experimental Setting

The experimental framework is designed to evaluate LLM-based subjectivity annotation in Spanish, utilizing a traditional machine learning pipeline as a baseline control. This section details the evaluation corpus (Section 5.1), outlines the curation of the human gold standard (Section 5.2), and describes the implementation of both the traditional control (Section 5.3) and the generative LLM configurations (Section 5.4). By relying on Part-of-Speech (PoS) tagging and handcrafted lexicons, this traditional control establishes a stable baseline to benchmark the nature and distribution of generative label noise.

5.1 Dataset

This study uses the *FakeNewsSpanishCorpus* (Posadas-Durán et al., 2019) and *RUN* (Bonet-Jover et al., 2023) datasets, both validated for Spanish disinformation tasks. The selection of these corpora is grounded in the correlation between speaker implication and the spread of unreliable content (Vieira et al., 2020; Jeronimo et al., 2019). These sources cover diverse text formats, including

traditional news, social media, blogs, and independent websites, to capture the varied communicative styles of User-Generated Content (UGC).

For the diagnostic analysis, a high-density evaluation corpus of 52,569 words across 339 sentences is extracted. Although compact, this sample size is a deliberate and statistically justified choice. Based on an estimated amount of 18,000 sentences in the source corpora, this 339-sentence sample yields a 5.3% margin of error at a 95% confidence level ($z = 1.96$, $p = 0.5$). At the token level, this constitutes over 13% of the source text. This scale ensures statistical representativeness while keeping detailed, word-level manual validation feasible. The corpus exhibits a significant annotation density, with nearly one in four words (23.23%) carrying a tagged element related to subjectivity.

5.2 Noise-Free Ground-Truth Standard

To evaluate automated label noise, a gold-standard reference was established through the manual annotation of the complete dataset by an expert linguist. While relying on a single annotator is a structural limitation, this choice is deliberate. Because the objective is to analyze error typologies rather than produce statistical generalizations, removing human inter-annotator variance isolates a highly consistent pragmatic baseline. This noise-free reference is essential to analyze and quantify systemic LLM deviations without introducing confounding variables.

Since this study’s objective is to diagnose models’ metalinguistic annotation capabilities, not human performance, the gold standard’s role is methodological: ensuring that discrepancies in model outputs can be attributed to model behavior rather than reference inconsistency.

5.3 Traditional Lexicon- and Rule-Based Baseline

To evaluate generative models against a traditional control, a rule- and lexicon-based pipeline was implemented using the SpaCy library (Honnibal et al., 2020). This system serves as an interpretable baseline to determine whether label noise stems from the pragmatic complexity of the text or from generative variability.

The pipeline processes Spanish text at the token level through three sequential phases: First, morphological pre-processing applies Part-of-Speech (PoS) tagging to map the grammatical structure. Second, lexicon-based matching targets semanti-

cally rich markers; this phase integrates verbs of knowledge and belief extracted from the ADESSE database (García-Miguel et al., 2010)¹, alongside generalization terms and temporal markers. Third, rule-based extraction implements custom heuristics to isolate complex structural patterns, such as standard passive constructions and the Spanish impersonal reflexive passive (utilizing the clitic ‘se’). Finally, overlapping outputs are deduplicated to establish a stable reference for error profiling.

Reporting these three components (PoS tagging, lexicon matching, rule-based extraction) as a unified traditional ML control, rather than in isolation, was a deliberate choice. A prior exhaustive study (Pérez-Montero et al., 2026) determined the optimal ML configuration for this annotation task, showing that spaCy’s PoS tagging alone was limited on complex, Spanish-specific phenomena, and that combining it with handcrafted lexicons and rules yielded the strongest performance. This combined pipeline was therefore adopted as the single traditional control, rather than reporting each component separately.

5.4 Evaluated Generative Models

To profile generative label noise across different architectural families, three state-of-the-art models were selected following the criteria of Beguš et al. (2025). These include two proprietary architectures accessed via paid API’s, **GPT-5** (OpenAI, 2025) and **Gemini 2.5 Pro** (Gemini Team, 2025), and one open-source model accessed locally via Ollama, **Llama 4 Maverick Instruct** (Team, 2025). For this diagnostic evaluation, all models were queried using default decoding parameters to capture their baseline noise distribution profiles.

5.5 Prompting Strategies and Annotation Task

The automated annotation is framed as a word- and span-level tagging task. Models are required to identify and extract the exact tokens or multi-word expressions that function as the active linguistic markers for each subjectivity dimension (as detailed in Table 1). Because the target corpus is in Spanish, both the system prompts and the model outputs are processed entirely in Spanish to pre-

¹The University of Vigo’s Database of Verbs, Diathesis Alternations, and Syntactic-Semantic Schemas of Spanish (*Base de datos de Verbos, Alternancias de Diátesis y Esquemas Sintáctico-Semánticos del Español*), available at: <https://adesse.webs.uvigo.es/>.

serve the pragmatic and syntactic nuances of the language.

Following the taxonomy of Liu et al. (2023) and the TELeR hierarchy (Karmaker Santu and Feng, 2023), three prompting configurations were engineered to examine how context layout triggers or mitigates omission and addition errors. All strategies share an identical system instruction that primes the models as expert linguists, isolating pragmatic reasoning from general task compliance. The configurations are structured as follows:

- **Decomposition (TELeR Level 2-3):** A direct, zero-shot directive targeting structural grammatical extractions.
- **Composition (TELeR Level 4):** A step-by-step Chain-of-Thought (CoT) structure enforcing explicit logical checkpoints before label decision.
- **Augmentation (TELeR Level 4-5):** A few-shot learning framework fueled with exemplary context-label sentences.

The full Spanish prompt templates and annotation examples are provided in Appendix B. Since this work is an exploratory analysis, one of whose objectives is to evaluate models’ metalinguistic capacity, the complete annotation guidelines were not provided in the prompts. Instead, different prompts were tailored to analyze how models annotate subjectivity under different prompting approaches with varying levels of information, following the Liu et al. (2023) taxonomy. The human annotator followed the full annotation guidelines, as they were also its author; these guidelines are included in Appendix A only to aid readability and are not directly comparable to the prompts used for the models.

6 Diagnostic Results and Noise Analysis

To characterize generative label noise, all automated annotations are evaluated against the human gold standard (Section 5.2). This analysis is structured across three diagnostic layers. First, a macro-level evaluation uses Precision, Recall, and F1-score to establish an overall performance baseline (Subsection 6.1). Second, a granular analysis maps the density of omission and addition errors across each subjectivity dimension (Figure 1). Finally, a qualitative characterization examines how specific linguistic features trigger systematic label noise in both the traditional control and the generative configurations (Subsection 6.2).

6.1 Macro-Level Quantitative Analysis

Within this initial evaluation, **Precision** measures the mitigation of addition errors (false positives), **Recall** reflects the avoidance of omission errors (false negatives), and the **F1-score** measures the overall alignment. These baseline results across all configurations are detailed in Table 3.

Automatic Annotation Noise Profiles. The results reveal distinct label noise patterns across the evaluated systems. The traditional ML control minimizes addition errors (achieving high precision) in low-commitment (LC) categories, such as *Doubt* ($F_1 = 0.78$), *Non-specificity* ($F_1 = 0.80$), and *Distancing* ($F_1 = 0.88$). Conversely, generative LLMs reduce omission errors (yielding higher recall) in high-commitment (HC) categories, such as *Certainty*, *Specificity*, and *Participation*. Among these, Gemini minimizes overall noise, exceeding an F_1 of 0.89 across all HC dimensions while balancing additions and omissions. GPT configurations yield higher systematic noise, peaking at a lower F1-score of 0.63 in *Specificity*. Llama effectively controls omissions in *Specificity* ($F_1 = 0.94$ under the Composition prompt), though it generally produces more addition noise than Gemini.

Impact of Prompt Configurations on Noise Stability. Prompt engineering affects noise generation unevenly across architectures. While GPT and Gemini exhibit stable alignment profiles (with F1 variance of $\Delta F_1 < 0.10$ and < 0.20 , respectively), Llama shows high susceptibility to prompt variations. In *Participation* and *Distancing*, Llama’s F1-scores fluctuate by up to 0.46, demonstrating that prompt formulation heavily dictates performance stability in open-source models.

Statistical Validation of Noise Distribution. Dimensions with concrete structural markers, such as *Specificity*, consistently generate lower label noise. This suggests that explicit linguistic elements (e.g., named entities) are more resilient to automated error. To validate these patterns, a Friedman test (Friedman, 1937) was applied to the F_1 results. The test confirms that performance variations across models, prompts, and categories are statistically significant ($Q = 31.85, p < 0.001$), supporting the need for a granular qualitative diagnostic of these error patterns.

Model_Configuration	Doubt (LC)			Non-specificity (LC)			Distancing (LC)		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
Traditional ML Control	0.72	0.84	0.78	0.77	0.84	0.80	0.88	0.88	0.88
GPT_Decomposition	0.31	0.40	0.35	0.30	0.41	0.35	0.37	0.49	0.43
GPT_Composition	0.28	0.32	0.30	0.28	0.35	0.31	0.36	0.38	0.37
GPT_Augmentation	0.32	0.37	0.34	0.14	0.25	0.18	0.32	0.37	0.34
Gemini_Decomposition	0.40	0.53	0.46	0.47	0.80	0.59	0.39	0.99	0.56
Gemini_Composition	0.43	0.56	0.49	0.47	0.81	0.59	0.51	0.97	0.67
Gemini_Augmentation	0.43	0.55	0.48	0.49	0.79	0.61	0.65	0.90	0.76
Llama_Decomposition	0.31	0.40	0.35	0.26	0.61	0.37	0.25	0.32	0.28
Llama_Composition	0.20	0.25	0.22	0.32	0.82	0.46	0.33	0.72	0.46
Llama_Augmentation	0.23	0.30	0.26	0.18	0.77	0.30	0.69	0.77	0.73

Model_Configuration	Certainty (HC)			Specificity (HC)			Participation (HC)		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
Traditional ML Control	0.46	0.48	0.47	0.69	0.84	0.76	0.92	0.93	0.93
GPT_Decomposition	0.31	0.40	0.35	0.57	0.58	0.58	0.47	0.51	0.49
GPT_Composition	0.28	0.39	0.33	0.62	0.63	0.63	0.49	0.52	0.50
GPT_Augmentation	0.33	0.46	0.39	0.59	0.61	0.60	0.41	0.44	0.42
Gemini_Decomposition	0.78	0.85	0.81	0.94	0.99	0.97	0.95	0.95	0.95
Gemini_Composition	0.84	0.88	0.86	0.94	0.97	0.96	0.96	0.96	0.96
Gemini_Augmentation	0.87	0.90	0.89	0.80	0.99	0.88	0.93	0.94	0.94
Llama_Decomposition	0.32	0.45	0.37	0.81	0.91	0.86	0.70	0.72	0.71
Llama_Composition	0.22	0.28	0.25	0.92	0.95	0.94	0.59	0.62	0.61
Llama_Augmentation	0.26	0.43	0.32	0.75	0.93	0.83	0.20	0.36	0.25

Table 3: Noise susceptibility benchmarks (Precision, Recall, and F1-score) across the six multi-dimensional scales. Bold values indicate optimal configuration boundaries against synthetic label noise injection.

6.2 Qualitative Analysis of Error Profiles

To analyze the linguistic roots of automated label noise, individual model errors are mapped directly to the proposed diagnostic taxonomy (Section 4). This qualitative profiling isolates specific instances of omission and addition errors.

Table 4 shows divergent error profiles across the evaluated systems. The traditional ML control generates errors predominantly through omissions (63.99%). In contrast, generative models exhibit a pronounced tendency toward addition errors. Within the LLM configurations, Gemini and Llama are heavily biased toward additions (exceeding 51%), whereas GPT produces a high rate of omissions (exceeding 70%) alongside lower overall match rates.

To examine these error patterns across specific linguistic contexts, Table 5 details the distribution at the dimension level, using the optimal prompt configuration for each model family (Composition for GPT and Llama; Augmentation for Gemini).

Low-Commitment Dimensions (LC). This category groups the dimensions where speaker stance is attenuated or references are detargeted: *Doubt*, *Non-specificity*, and *Distancing*.

- **Doubt:** This dimension generates high omis-

Experimental Setup	Noise Type (Relative %)			
	% Match	% Error	% Omit	% Add
Traditional ML Control	72.26	27.74	63.99	36.01
GPT_Decomposition	41.50	58.50	72.81	27.19
GPT_Composition	42.36	57.64	79.20	20.80
GPT_Augmentation	38.78	61.22	70.97	29.03
Gemini_Decomposition	63.96	36.04	18.00	82.00
Gemini_Composition	69.84	30.16	23.86	76.14
Gemini_Augmentation	71.81	28.19	28.69	71.31
Llama_Decomposition	48.60	51.40	48.35	51.65
Llama_Composition	48.84	51.16	35.09	64.91
Llama_Augmentation	42.31	57.69	28.12	71.88

Table 4: Global distribution of label noise across experimental configurations. Noise types describe the relative ratio of omission (% Omit) versus addition (% Add) errors calculated over the total noise (Error) (100%).

sion noise in generative models. GPT and Llama produce substantial overall error (72.16% and 79.59%, respectively), primarily driven by omissions (81.43% in GPT). While the traditional control yields a balanced 50/50 distribution, LLMs frequently overlook modal markers (e.g., the Spanish subjunctive) unless explicit lexical indicators are present.

- **Non-specificity:** Polysemous indefinite terms often trigger addition errors in LLMs. While the traditional control achieves a 76.53%

Optimized Setup	% Match	% Error	Noise Type (Relative %)	
			% Omit	% Add
Doubt (LC)				
Traditional ML Control	72.48	27.52	50.00	50.00
GPT_Composition	27.84	72.16	81.43	18.57
Gemini_Augmentation	42.96	57.04	60.49	39.51
Llama_Composition	20.41	79.59	76.92	23.08
Certainty (HC)				
Traditional ML Control	46.04	53.96	94.04	5.96
GPT_Composition	28.04	71.96	60.00	40.00
Gemini_Augmentation	87.11	12.89	74.68	25.32
Llama_Composition	22.27	77.73	73.28	26.72
Non-specificity (LC)				
Traditional ML Control	76.53	23.47	61.96	38.04
GPT_Composition	28.39	71.61	72.52	27.48
Gemini_Augmentation	49.20	50.80	25.09	74.91
Llama_Composition	31.68	68.32	10.09	89.91
Specificity (HC)				
Traditional ML Control	68.99	31.01	42.47	57.53
GPT_Composition	62.47	37.53	96.05	3.95
Gemini_Augmentation	80.33	19.67	3.76	96.24
Llama_Composition	91.76	8.24	54.44	45.56
Distancing (LC)				
Traditional ML Control	88.36	11.64	100.00	0.00
GPT_Composition	35.52	64.48	91.10	8.90
Gemini_Augmentation	65.06	34.94	20.65	79.35
Llama_Composition	33.26	66.74	19.12	80.88
Participation (HC)				
Traditional ML Control	92.44	7.56	94.12	5.88
GPT_Composition	48.84	51.16	87.50	12.50
Gemini_Augmentation	93.33	6.67	86.67	13.33
Llama_Composition	59.34	40.66	87.84	12.16

Table 5: Granular noise diagnostics partitioned by dimension under optimal prompting configurations. Noise types describe the relative ratio of omission (% Omit) versus addition (% Add) errors.

match rate, Llama underperforms, generating 89.91% of its errors as additions. High-frequency indefinites (e.g., *todo*, *alguien*) act as noise triggers, causing models to over-annotate without sufficient context analysis.

- **Distancing:** The Spanish impersonal reflexive (*pasiva refleja* with the clitic ‘*se*’) reveals contrasting behaviors. The traditional control incurs zero additions, maintaining a low overall error rate (11.64%). Conversely, Gemini Augmentation yields 79.35% addition errors, over-interpreting the grammatical marker ‘*se*’ as a subjective perspective marker even in objective contexts.

High-Commitment Dimensions (HC). This category encompasses dimensions featuring assertive speaker commitment and explicit anchoring: *Certainty*, *Specificity*, and *Participation*.

- **Certainty:** Distinct error dynamics emerge

here. The traditional control is highly conservative, showing almost exclusive omission noise (94.04%) and nearly zero additions (5.96%). Gemini Augmentation minimizes overall noise to 12.89%, but shifts the bias toward additions (25.32%). This indicates that mitigating omissions in LLMs often introduces synthetic addition noise by over-annotating factual sentences as certain.

- **Specificity:** Explicit syntactic markers (e.g., named entities, numbers) suppress omission noise. Llama Composition achieves the lowest overall noise density (8.24%). However, the residual noise in Gemini Augmentation is almost entirely composed of additions (96.24%), reflecting a systematic over-annotation of objective referential tokens.
- **Participation:** Gemini Augmentation achieves the highest overall alignment, limiting total noise to 6.67%. The error profiles are highly polarized: the traditional control and GPT generate noise almost exclusively via omissions (94.12% and 87.50%, respectively), whereas Gemini balances additions and omissions, demonstrating a robust capability to process explicit personal pronouns and verbal inflections.

Overall Noise Bias Analysis. Figure 1 reports the error bias distribution for each model configuration and dimension, computed as the normalized difference between omission and addition error proportions, ranging from +1 (all errors are omissions) to −1 (all errors are additions), with values near 0 indicating a balanced mix. The traditional control and GPT cluster toward positive values across most dimensions (e.g., traditional control: +1.00 in *Distancing* and *Participation*; GPT_Augmentation: +1.00 in *Doubt*), confirming a systematic failure to detect subjective indicators. Conversely, Gemini shifts toward negative values in *Specificity* (Gemini_Augmentation: −0.56) and *Distancing* (Gemini_Decomposition: −1.00), reflecting a tendency to over-generate fake markers, while Llama shows a more mixed profile in *Specificity*, ranging from −0.20 (Decomposition) to +0.71 (Composition). *Participation* shows the strongest and most consistent omission bias across generative models (+1.00 for Gemini across all three prompting strategies), while *Doubt* remains predominantly omission-biased across most setups, with Gemini as the exception, showing a mild ad-

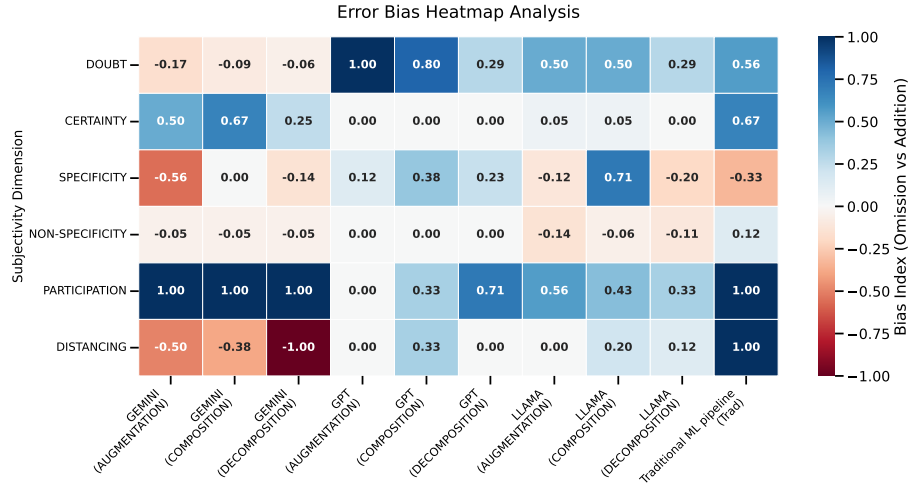


Figure 1: Error bias distribution across categories and model configurations. Positive values (blue) indicate a tendency toward omissions, while negative values (red/orange) indicate a tendency toward additions.

dition bias across all its configurations (-0.17 to -0.06).

7 Conclusion and Future Work

This study presents a diagnostic framework to analyze, categorize, and benchmark label noise in automated text annotation. Beyond standard macro-performance metrics, this approach demonstrates how contextual and grammatical ambiguity translates into systematic error profiles when employing LLMs as automated annotators.

The findings indicate distinct error patterns dictated by system architecture. While the traditional ML control predominantly generates omission errors, generative language models exhibit a pronounced tendency to introduce addition errors when annotating complex subjectivity dimensions. These discrepancies are particularly severe in Spanish structures, such as subjunctive alternations and the impersonal reflexive passive, where automated systems show decreased stability in label assignment. This instability introduces significant label noise, underscoring the necessity of human-in-the-loop validation before LLMs can be reliably adopted as automated annotators.

Taken together, the dimension-level analysis (Section 6) shows that models’ metalinguistic capacity is variable, not uniformly strong or weak. Performance improves on dimensions with explicit referential markers, such as numbers or named entities in *Specificity*, which are easy to identify regardless of context. In contrast, performance drops on dimensions that depend on more complex grammatical resolution, most notably *Doubt*, where cues

like the subjunctive mood are harder to detect reliably. Crucially, this same pattern appears in the traditional ML control, not only in LLMs, suggesting that the difficulty lies in the subjective nature of the annotation task itself rather than in the metalinguistic capacity of the models alone, given that annotation disagreement is not always an error but may reflect legitimate diversity in interpretation (Piot et al., 2026).

Consequently, the proposed taxonomy provides a structured methodology to audit automated dataset reliability. Future work will mainly focus on expanding this diagnostic approach to multi-annotator environments, as well as performing further analysis on the applicability of subjectivity-aware evaluation metrics that can handle the variability reported in previous work (Piot et al., 2026; Sharma et al., 2026). Additional research lines could include engineering prompt strategies specifically designed to mitigate addition errors, exploring repeated-querying annotation protocols instead of single-shot predictions to improve consistency on subjective tasks to generate coherent annotations that reduce noise (Sharma et al., 2026), and validating the stability of these synthetic noise profiles across other typologically diverse languages.

Limitations

This study presents three primary limitations. First, although the dataset size is constrained, this scale is a deliberate trade-off enabling the exhaustive, word-level manual analysis required to profile label noise. Conducting a manual validation of this granularity on substantially larger corpora would

be logistically unfeasible.

Second, the observed error distributions are closely tied to Spanish subjectivity markers (Klie et al., 2023), such as subjunctive alternations and the clitic “se”. While these specific patterns are language-dependent, the underlying diagnostic taxonomy remains highly generalizable and can be adapted to analyze label noise in other linguistic contexts.

Finally, this evaluation is limited to a specific snapshot of large language models and prompting strategies. While model updates will inherently alter absolute noise baselines over time, the proposed methodology establishes a reproducible framework to benchmark, track, and audit automated annotation workflows.

Ethical Considerations

This study utilizes exclusively public, anonymized, and benchmarked datasets, avoiding direct data collection. Additionally, this research focuses solely on characterizing automated label noise to improve dataset quality, without generating synthetic content or promoting harmful applications.

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A Annotation Scheme: Dimensions and Linguistic Markers

Table 6 presents the automatic annotation framework, outlining the linguistic indicators and examples evaluated for each dimension.

B Prompting Configuration Examples

This appendix presents the templates for the prompting configurations discussed in Section 5.5. To preserve the original contextual features, all instructions were executed in Spanish. The following examples use the *Doubt* dimension as a baseline reference to illustrate the structural variations across the evaluation settings.

- **Decomposition.** The instruction uses a direct, zero-shot directive for marker extraction.

<You are an expert linguist. Your goal is to extract words, sets of words, or punctuation marks that indicate DOUBT, as expressed verbally through Rhetorical Questions (question marks), Conditional Verbs, Subjunctive Mood Verbs (...). Analyze the following text and return the annotated text, marking the extracted text fragment using angle brackets '<>' and the dimension DOUBT using square brackets '[']. The text to analyze is as follows: [Insert text here].>

- **Composition.** The instruction uses a step-by-step chain-of-thought structure to guide the annotation process.

<You are an expert linguist. Your objective is to extract words, sets of words, or punctuation marks that indicate DOUBT; to achieve this, you must follow six steps: Step 1: Search for question marks (¿?); Step 2: Locate conditional verbs (e.g., “sería” [would be], “podría” [could]); Step 3: Detect subjunctive mood verbs (e.g., “dudara” [I/he/she doubted], “sospechara” [I/he/she suspected]); Step 4: Mark verbs of belief based on their meaning (e.g., “cuestionar” [to question], “desconfiar” [to mistrust]); Step 5: Extract interrogative pronouns (e.g., “qué” [what], “cómo” [how]); and finally, Step 6: Analyze the text and return the annotated text as output, marking the extracted text fragment using angle brackets '<>' and the dimension DOUBT using square brackets '['], utilizing the following text for analysis: [Insert text here].>

- **Augmentation.** The instruction includes in-context examples from the corpus to guide the automatic labeling process (few-shot learning).

<You are an expert linguist. Your objective is to annotate words, word groups, or punctuation marks that

Dimension	Linguistic Markers	Examples (Spanish / English)
Degree of Certainty		
<i>Doubt</i> (LC)	- Conditional and subjunctive - Modal adverbs (<i>quizá</i>) - Verbs of belief (<i>creer</i>)	- “<i>Quizá sea tarde</i>” (<i>Maybe it’s late</i>) - “<i>Creo que podría funcionar</i>” (<i>I think it could work</i>) - “<i>Parece que lloverá</i>” (<i>It seems it will rain</i>)
<i>Certainty</i> (HC)	- Affirmative adverbs (<i>sí</i>) - Future tense (conviction) - Verbs of knowledge	- “<i>Sé que vendrá</i>” (<i>I know he/she will come</i>) - “<i>Es evidente el cambio</i>” (<i>The change is evident</i>) - “<i>Esto garantizará el éxito</i>” (<i>This will guarantee success</i>)
Level of Concreteness		
<i>Non-specificity</i> (LC)	- Generalizing terms - Indefinite pronouns - Vague quantifiers	- “<i>Siempre dicen lo mismo</i>” (<i>They always say the same</i>) - “<i>Alguien debería actuar</i>” (<i>Someone should act</i>) - “<i>Todo el mundo lo sabe</i>” (<i>Everyone knows it</i>)
<i>Specificity</i> (HC)	- Proper names and acronyms - Specific dates - Technical entities	- “<i>Hablamos con García</i>” (<i>We spoke with García</i>) - “<i>Ocurrió en Madrid</i>” (<i>It happened in Madrid</i>) - “<i>El informe de la ONU</i>” (<i>The UN report</i>)
Scale of Involvement		
<i>Distancing</i> (LC)	- Impersonal reflexive (<i>se</i>) - Passive voice - 3rd person plural agency	- “<i>Se dice que...</i>” (<i>It is said that...</i>) - “<i>Fue decidido ayer</i>” (<i>It was decided yesterday</i>) - “<i>Comentan que hay problemas</i>” (<i>They say there are problems</i>)
<i>Participation</i> (HC)	- 1st/2nd person pronouns - Possessives - Direct address	- “<i>Mi opinión es...</i>” (<i>My opinion is...</i>) - “<i>Queremos informarte</i>” (<i>We want to inform you</i>) - “<i>¿Qué opinas tú?</i>” (<i>What do you think?</i>)

Table 6: Annotation framework categorized by dimension, linguistic markers, and translated examples.

indicate *DOUBT*, as expressed verbally through question marks, Conditional Verbs, Subjunctive Mood Verbs, (...). The reference examples are: 1. “*Si fuera natural, solo habría afectado a países con la misma temperatura que China*” (where ‘*Si*’ [*If*] indicates conditional subordination, ‘*fuera*’ [*were*] is subjunctive, and ‘*habría*’ [*would have*] is conditional) (...). Perform the text analysis and return the annotated text as output, marking the extracted text fragment using angle brackets ‘<>’ and the dimension *DOUBT* using square brackets ‘[]’, utilizing the following text for analysis: [Insert text here].>